



Solution Set

Question 1

Given prices 100, 97, 101, 99, 104, 102, and 108, compute the six arithmetic returns and six log returns. Which return type aggregates exactly over time?



Answer key

Arithmetic returns are computed as

$$r_t = \frac{S_t - S_{t-1}}{S_{t-1}}$$

(−3.0000%, 4.1237%, −1.9802%, 5.0505%, −1.9231%, 5.8824%)

Log returns are computed as

$$\ell_t = \log\left(\frac{S_t}{S_{t-1}}\right)$$

(−3.0459%, 4.0411%, −2.0001%, 4.9271%, −1.9418%, 5.7143%)

Log returns aggregate exactly because

$$\sum_{t=1}^6 \ell_t = \log\left(\frac{108}{100}\right) = 7.6961\%$$

Question 2

Use the returns in Question 1. Compute the sample mean and sample volatility of the arithmetic returns.

Answer key

Using decimal returns, the sample mean is

$$\bar{r} = \frac{1}{6} \sum_{t=1}^6 r_t = 0.013589$$

or 1.3589 percent. The sample variance uses n-1 in the denominator:

$$s^2 = \frac{1}{5} \sum_{t=1}^6 (r_t - \bar{r})^2 = 0.001550$$

Therefore the sample volatility is

$$s = \sqrt{0.001550} = 3.937\%$$

In risk work, state whether volatility is daily, weekly, monthly, or annualized.

Question 3

Annualize the daily volatility from Question 2 using 252 trading days. State the assumption behind this scaling.

Answer key

The usual square-root-of-time annualization is

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \sqrt{252}$$
$$0.03937 \sqrt{252} = 0.6250$$

so annualized volatility is about 62.50 percent. This assumes returns are approximately uncorrelated with stable daily variance; it can fail under volatility clustering or regime shifts.

Question 4

For an AR(1) model

$$X_t = c + \phi X_{t-1} + \epsilon_t$$

derive the unconditional mean and variance under stationarity.

Answer key

Taking expectations under stationarity gives

$$\mu = c + \phi \mu \Rightarrow \mu = \frac{c}{1-\phi}$$

Subtract the mean:

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t$$

Taking variances and assuming innovations are independent of the past:

$$\gamma_0 = \phi^2 \gamma_0 + \sigma_\epsilon^2 \Rightarrow \gamma_0 = \frac{\sigma_\epsilon^2}{1-\phi^2}$$

These formulas require

$$|\phi| < 1$$

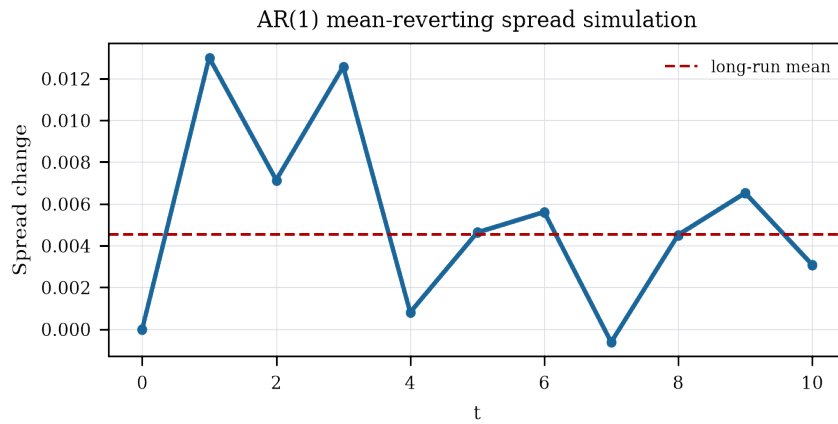
Question 5

An AR(1) spread model has

$$c = 0.001, \quad \phi = 0.78$$

and innovation volatility 1.20 percent. Compute the long-run mean, long-run volatility, and next forecast when

$$X_t = 0.030$$



Answer key

The long-run mean is

$$\mu = \frac{0.001}{1-0.78} = 0.004545$$

The unconditional variance is

$$\frac{0.012^2}{1-0.78^2} = \frac{0.000144}{0.3916} = 0.0003677$$

so long-run volatility is

$$\sqrt{0.0003677} = 1.918\%$$

The one-step forecast is

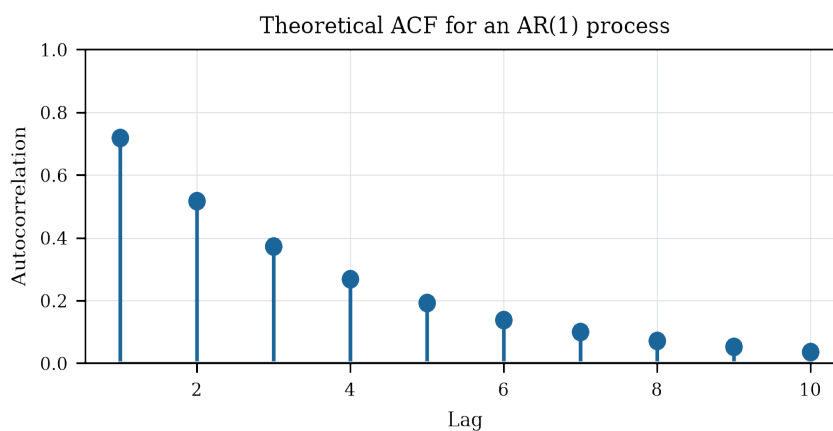
$$\mathbb{E}[X_{t+1} | X_t = 0.030] = 0.001 + 0.78(0.030) = 0.0244$$

Question 6

For an AR(1) with

$$\phi = 0.72$$

compute the theoretical autocorrelations at lags 1, 3, and 10. Interpret the plotted decay.



Answer key

For a stationary AR(1), the lag-k autocorrelation is

$$\rho_k = \phi^k$$

Therefore

$$\rho_1 = 0.72, \quad \rho_3 = 0.72^3 = 0.3732, \quad \rho_{10} = 0.72^{10} = 0.0374$$

The autocorrelation decays geometrically. Slow decay near one is a warning sign for high persistence or possible nonstationarity.

Question 7

A time series follows

$$X_t = X_{t-1} + \epsilon_t$$

with

$$\text{Var}(\epsilon_t) = \sigma^2$$

Derive

$$\text{Var}(X_t)$$

when

$$X_0 = 0$$

Why is this a unit-root problem?

Answer key

Iterating the recursion gives

$$X_t = \sum_{i=1}^t \epsilon_i$$

With independent innovations,

$$\text{Var}(X_t) = \sum_{i=1}^t \text{Var}(\epsilon_i) = t\sigma^2$$

Variance grows with time instead of settling to a finite long-run value. That is the practical danger of a unit root: shocks have permanent effects.

Question 8

Derive the OLS estimator for

$$y = X\beta + u$$

and state the exogeneity condition needed for unbiasedness.

Answer key

OLS minimizes

$$S(\beta) = (y - X\beta)^\top (y - X\beta)$$

Differentiate and set the gradient to zero:

$$\nabla_{\beta} S = -2X^\top (y - X\beta) = 0$$

$$X^\top X\hat{\beta} = X^\top y$$

If $X^\top X$ is invertible,

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

A key condition for unbiasedness is

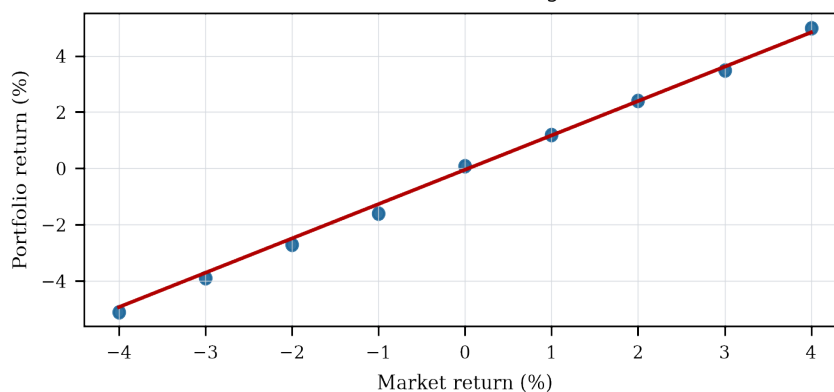
$$\mathbb{E}[u | X] = 0$$

Question 9

Using the scatter plot, estimate whether the portfolio has positive market beta. Then compute predicted portfolio return for a market return of -3 percent if

$$\hat{r}_p = -0.05\% + 1.22r_m$$

Portfolio return beta regression



Answer key

The fitted line slopes upward, so the market beta is positive. For a -3 percent market return,

$$\hat{r}_p = -0.05\% + 1.22(-3\%)$$

$$\hat{r}_p = -0.05\% - 3.66\% = -3.71\%$$

The regression converts a market-factor shock into a conditional portfolio-return forecast.

Question 10

A portfolio return regression has

$$\hat{\alpha} = 0.0002, \quad \hat{\beta} = 1.4$$

market volatility 1.5 percent, and residual volatility 0.8 percent. Decompose variance into systematic and idiosyncratic parts.

Answer key

In a one-factor regression,

$$r_p = \alpha + \beta r_m + \epsilon$$

the model-implied variance is

$$\text{Var}(r_p) = \beta^2 \sigma_m^2 + \sigma_\epsilon^2$$

Systematic variance is

$$1.4^2(0.015^2) = 0.000441$$

Residual variance is

$$0.008^2 = 0.000064$$

Total variance is 0.000505, so about 87.3 percent of variance is systematic.

Question 11

Explain omitted-variable bias in a factor-risk regression and give the one-omitted-factor formula.

Answer key

Suppose the true model is

$$y = \beta_1 x_1 + \beta_2 x_2 + u$$

but x_2 is omitted. If x_1 and x_2 are correlated, the included beta absorbs part of the missing factor:

$$\text{plim } \hat{\beta}_1 = \beta_1 + \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}$$

This is critical in market risk because missing a correlated factor can make a hedge look better or worse than it really is.

Question 12

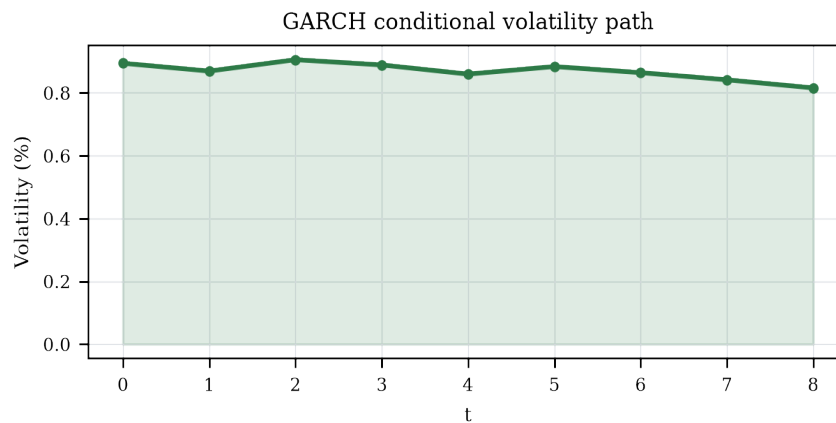
A GARCH(1,1) model is

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

For

$$\omega = 0.000002, \quad \alpha = 0.08, \quad \beta = 0.90$$

check stationarity and compute long-run volatility.



Answer key

Covariance stationarity requires

$$\alpha + \beta < 1$$

Here

$$0.08 + 0.90 = 0.98 < 1$$

so the process is stationary but highly persistent. Long-run variance is

$$\frac{\omega}{1 - \alpha - \beta} = \frac{0.000002}{0.02} = 0.0001$$

so long-run volatility is

$$\sqrt{0.0001} = 1.00\%$$

Question 13

Using the GARCH model in Question 12, compute tomorrow's conditional volatility if

$$\epsilon_t = -2\%, \quad \sigma_t = 1.1\%$$

Answer key

The variance forecast is

$$\begin{aligned} \sigma_{t+1}^2 &= 0.000002 + 0.08(0.02^2) + 0.90(0.011^2) \\ &= 0.000002 + 0.000032 + 0.0001089 = 0.0001429 \end{aligned}$$

Therefore

$$\sigma_{t+1} = \sqrt{0.0001429} = 1.195\%$$

The large squared shock raises tomorrow's volatility forecast.

Question 14

A 99 percent VaR model has 8 exceptions in 250 days. Compute the expected number of exceptions and give the likelihood of exactly 8 exceptions under the correct model.

Answer key

Under correct unconditional coverage, exceptions follow approximately

$$N \sim \text{Binomial}(250, 0.01)$$

Expected exceptions are

$$\mathbb{E}[N] = 250(0.01) = 2.5$$

The probability of exactly 8 exceptions is

$$\binom{250}{8} (0.01)^8 (0.99)^{242} \approx 0.0081$$

Eight exceptions is therefore a serious warning sign, though a full test usually considers eight or more exceptions.

Question 15

For two nonstationary price-level series, explain why a high R-squared regression can be spurious. What would you check before using the relationship?



Answer key

Trending series can move together visually even when there is no stable economic relationship. A regression in levels can then show high R-squared and significant t-statistics by accident. Before using it, check stationarity, inspect residuals, test for unit roots, and test whether a spread such as

$$Y_t - \theta X_t$$

is stationary. If the spread is stationary, the series may be cointegrated.

Question 16

Define cointegration. If

$$Y_t - 1.3X_t$$

is stationary while X and Y are each

$$I(1)$$

what does that imply for relative-value risk?

Answer key

Cointegration means nonstationary series have a stationary linear combination:

$$Y_t - \theta X_t \sim I(0)$$

Here $\theta = 1.3$. For relative-value risk, the spread may be modeled as mean-reverting, but the individual legs remain nonstationary. The hedge ratio should be treated as an estimated relationship, not a permanent law.

Question 17

A residual series from a factor model has sample autocorrelation

$$\hat{\rho}_1 = 0.32$$

with

$$n = 100$$

Use the rough standard error

$$1/\sqrt{n}$$

to assess whether residual autocorrelation is material.

Answer key

The rough standard error under no autocorrelation is

$$\frac{1}{\sqrt{100}} = 0.10$$

The observed autocorrelation is about 3.2 standard errors from zero:

$$\frac{0.32}{0.10} = 3.2$$

That is material. It suggests the model residuals still contain time-series structure and the factor model may be missing dynamics.

Question 18

A regression estimated on daily returns gives residuals with volatility 1 percent and clear volatility clustering. Why are usual OLS standard errors suspect, and what would you use instead?

Answer key

OLS coefficients can still be useful under some conditions, but usual standard errors assume homoscedastic and often uncorrelated errors. Volatility clustering violates that assumption. A standard response is to use heteroscedasticity-robust or HAC standard errors, and for forecasting risk, model conditional variance explicitly with EWMA or GARCH.

Question 19

A 60-day EWMA volatility estimate uses

$$\lambda = 0.94$$

yesterday's variance

$$\sigma_t^2 = 0.0001$$

and today's squared return

$$r_t^2 = 0.0004$$

Compute tomorrow's variance forecast.

Answer key

EWMA updates variance as

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) r_t^2$$

Substitute the numbers:

$$\sigma_{t+1}^2 = 0.94(0.0001) + 0.06(0.0004) = 0.000118$$

The volatility forecast is

$$\sqrt{0.000118} = 1.086\%$$

Question 20

A two-factor regression has estimated betas

$$\beta_{\text{EQ}} = 1.1, \quad \beta_{\text{IR}} = -4.5$$

where the rate factor is measured in decimal yield changes. Estimate PnL impact for equity return -2 percent and rate move +5 bps.

Answer key

Convert 5 bps to decimal yield change:

$$5\text{bps} = 0.0005$$

The fitted PnL or return impact is

$$1.1(-0.02) + (-4.5)(0.0005) = -0.022 - 0.00225 = -0.02425$$

So the predicted impact is -2.425 percent. The sign of the rate beta means the portfolio loses when rates rise.